Tashkent Institute of Textile and Light Industry
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November 5, 2026

Editor-in-Chief
IEEE Internet of Things Journal

Dear Editor-in-Chief,

We submit the manuscript *"An IoT-Enabled Hybrid Soft Sensor for Real-Time Residual-Moisture Monitoring and Resource Accounting in Compact Wool Scouring Machines"* for consideration as a **Regular Paper**. We request **Traditional (non-open-access) publication**.

The work addresses IoT application design and IoT trials in an industrial setting that is currently not instrumented: wool scouring in small regional plants. Its contributions are (i) a low-cost ESP32-based architecture with a captive set-up portal, HTTP/MQTT transport, server analytics and a bilingual web/mobile application; (ii) a hybrid soft sensor for residual moisture that uses a design-of-experiments model of the squeezing device as a prior, with a staged, offset-shrinking online calibration from sparse laboratory samples; (iii) per-kilogram energy and water indicators and alarm rules with debounce and hysteresis; and (iv) three clearly separated layers of evidence: a reproducible reanalysis of historical mechanical-process data, a simulation and software bench study, and a factory validation [TODO: describe the completed field campaign: N shifts at plant, held-out accuracy, delivery ratio, latency, kWh/kg, L/kg]. The simulation reports openly where the prior helps (cold start, few labels, geometry changes) and where it does not (many labels at a fixed geometry). Code, firmware and analysis scripts are public at https://github.com/nasin0764-bit/woolwash-iot.

The manuscript is original, has not been published previously and is not under consideration elsewhere. [TODO: If any part was presented at a conference, cite it here and explain the extension.] [TODO: Insert the next sentence ONLY after every author has read and approved the complete manuscript:] All authors (R. Tuguzbayeva, S. Khakimov, F. Dustova, R. Rosulov and M. Anorkulov) have approved the submission and declare no conflicts of interest. [TODO: Author contributions: confirm with each author. Proposed for M. Anorkulov (pending his confirmation): statistical reanalysis of the factorial experiment, regression coefficients and RMSE/MAE/R^2.]

Suggested reviewers (no joint work with the authors): [TODO: Candidates are authors of cited work. Verify the current affiliation and e-mail of each, and that there is no conflict of interest, before submission:]

1. Zhiqiang Ge, soft sensors / industrial data analytics (cited: Sun and Ge, IEEE TII 2021) [TODO: affiliation, e-mail]
2. Wei Yu or Tharam Dillon, IoT monitoring in smart factories (cited: IEEE IoT-J 2022, IEEE TII 2023) [TODO: affiliation, e-mail]
3. Isabella Bianco, life-cycle assessment of wool processing (cited: J. Clean. Prod. 2023) [TODO: affiliation, e-mail]

Sincerely,

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